

Maximum Nullity of Cyclic Covers of Graphs

A monodromy ceiling, and the zero forcing
number of $P(n, k)$

Research Binder

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Overview & Standards of Proof

This research binder contains a dated record of the investigation into exact zero forcing numbers and maximum nullity of generalized Petersen graphs $P(n, k)$ and general cyclic covers.

Every claim in the paper is labelled by the standard it meets: **proved**, **proved by an exact certificate**, **proved by interval certification**, or **numerical**. Claims that have been refuted or withdrawn, including the author's own, are listed explicitly in the paper's Status section and dated in the log.

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I. Official Abstract

(September 28, 2026)

Fix the adjacency matrix of the generalized Petersen graph $P(n, k)$: is its spectral gap optimal? That is the Riemann Hypothesis for its Ihara zeta function, the Ramanujan condition. Let it range over every matrix with that pattern: how degenerate can zero be? That is the zero forcing number $Z(G)$, which bounds maximum nullity $M(G)$.

Fixed matrix: squaring twice, each step an equivalence, leaves one integer polynomial D_k . Substituting $x = \cos(j(k+1)/n)$, $y = \cos(jk/n)$ collapses it to $Q = 4x^2 + 4y^2 - 8|xy| + 3 = 0$, with no k in it: one fixed forbidden set, 0.43% of the square. Failure is then a Diophantine condition, forced for every k once $n \geq 231$ by Dirichlet's theorem: the family is finite in both variables. For $k \geq 9$ other than 6 and 8, Sturm root isolation of D_k proves the classification outright: $P(n, k)$ is Ramanujan iff n lies below an explicit bound, halved for odd n when k is odd. The finite remainder gives 460 pairs (n, k) , 324 graphs; none with $k > 45$. No infinite Ramanujan family is a cyclic cover.

Ranging matrix: eliminating inner coordinates leaves one recurrence of order $2k+2$, a ceiling equal to the known upper bound. A tiling theorem reduces all large n to finitely many identity tiles, verified in exact arithmetic; this gives $Z = M = 2k+2$ for all $n \geq 17$ at $k = 3$ and $n \geq 29$ at $k = 4$, determines $Z(P(n, k))$ for every n at $k = 4$, and proves Krishnan's Conjecture 5.

II. Research Plan

(September 28, 2026)

Project. Maximum Nullity of Cyclic Covers of Graphs: A Monodromy Ceiling and the Zero Forcing Number of Generalized Petersen Graphs **Category.** Mathematics **Finalist.** Aryan Padarathi

A. Rationale

Zero forcing began in two unrelated places: controllability of quantum spin systems, and placement of phasor measurement units in electrical grids. It was then connected to linear algebra by the inequality $M(G) \geq Z(G)$, where $M(G)$ is the largest nullity attainable by a symmetric matrix whose off-diagonal support is the edge set of G . The inequality makes Z a combinatorial ceiling on a purely algebraic quantity, and makes lower bounds on Z obtainable by exhibiting a matrix.

The generalized Petersen graphs $P(n, k)$ are a standard cubic test family. Rashidi, Shajareh Poursalavati and Tavakkoli (2020) proved $Z(P(n, k)) \geq 2k+2$ and determined $Z(P(n, 2))$. Their Theorem 3.6, asserting $Z(P(n, 3)) = 8$ for $n \geq 12$, is false: Krishnan's July 2026 correction note (arXiv:2607.19412) exhibits $Z(P(12, 3)) = 7$, the error being a case analysis that never excludes 7-vertex sets. That note leaves open the stabilization threshold and any rigorous lower bound for $k \geq 4$.

This project addresses those two open problems.

B. Research question

1. Is there an intrinsic ceiling on what a matrix certificate can prove about $Z(P(n, k))$, and where does it sit relative to the known upper bound?
2. Can that ceiling be attained, and by an explicitly checkable matrix?
3. Can $Z(P(n, k))$ be determined for **all** n rather than for sparse families of n ?

C. Procedures

This is a theoretical mathematics project. There are no human participants, no vertebrate animals, no potentially hazardous biological agents, and no hazardous chemicals, activities or devices. All work is proof and computation performed by the finalist on a personal computer.

C1 Structure. Eliminate the inner coordinates from the kernel equations of an arbitrary matrix carrying the $P(n, k)$ pattern and identify the resulting scalar recurrence, its order, and the leading coefficients. Establish when the nullity equals the recurrence order.

C2 Construction. For rotation-invariant matrices, reduce singularity to a polynomial condition on a discrete grid and construct matrices meeting the ceiling. Determine exactly when such a matrix is admissible.

C3 Obstruction. Determine whether the divisibility conditions arising in C2 are intrinsic, and prove the answer rather than observing it.

C4 General n . Construct matrices from repeated blocks whose contributions compose, so that one finite family of blocks settles an infinite set of n .

C5 Certification. Verify every constructed matrix at the highest standard available: exact integer or rational arithmetic where possible, and otherwise an interval (Krawczyk) contraction proof that a true real solution exists. Record which standard each claim meets.

C6 Independent checking. Compute Z directly by exhaustive search over vertex subsets for all sizes reachable, validate that solver against every published value, and reproduce each theorem numerically before relying on it.

D. Data analysis

Results are theorems and explicit matrices, not measurements, so analysis is verification rather than statistics. Every computational claim is re-derived by a single script that reports pass or fail per check. Numerical searches are always

wrapped in a bound they cannot legally exceed (the ceiling of C1), so that an impossible answer is detected rather than believed. Where a search fails, the residual and the computational budget are both recorded, because a failure at a small budget is not evidence of impossibility.

E. Risk and safety

None. No laboratory work, no fieldwork, no subjects of any kind.

F. Bibliography

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III. Short Note and Status of Claims

Placeholder: the student's own short note goes here, 4–6 pages, written from scratch. The paper's Status-of-claims section follows this page, taken unedited from the full paper. It states, for every claim, which of four standards it meets, and lists every claim of the author's own that has been refuted or withdrawn.

Part I is a complete classification of a recognised kind. Droll did this for unitary Cayley graphs and Le–Sander [12] for integral circulants of prime power order; in both cases, as here, only finitely many members qualify. The contribution is exactness and completeness, not surprise: anyone who knows that abelian covers do not expand would have predicted the shape of the answer. What was not available before is *which* members, and that no member exists beyond $k = 45$. For $k \leq 9$ other than 6 and 8 the answer is a theorem rather than a list (Theorem 25): the classification is an initial segment, cut at B_k , and at $B_k/2$ for odd n when k is odd, because the negative set of D_k consists of the end bands alone — a fact about the real roots of one integer polynomial, decided by Sturm’s theorem. Where the list has gaps, it is because an interior band appears, and from $k = 10$ on it always does.

Three things we are careful not to claim. First, the graph Riemann Hypothesis is not the classical one: by the Ihara determinant formula it is equivalent to a finite eigenvalue check, which is exactly why it is decidable. The analogy lives in the zeta function, not in the difficulty. Second, Theorem 30 refines a known principle rather than establishing a new one; what is new is the explicit constant and the exact criteria of Theorems 27 and 28 for the two-vertex bases. Third, the verification apparatus is methodology, not mathematics: it is why we believe the numbers, and it is not a theorem.

24 Status of the claims

Every claim below is labelled by the standard it meets. Four standards appear, in decreasing strength: *proved*, meaning a proof in the text; *proved by an exact certificate*, meaning an explicit matrix whose nullity is computed in $\mathbb{Z}$ or $\mathbb{Q}(\zeta_n)$; *proved by interval certification*, meaning a Krawczyk contraction bound establishing that a true real solution exists in an explicit box; and *numerical*, meaning a floating-point computation only. We also list what we have refuted, including our own earlier claims.

- **Proved.** Theorems 36, 40 and Theorem 44 with Corollary 45; Propositions 46 and 47; Theorem 63; Theorem 64 and Theorem 60, the nullity classifications of the adjacency matrix and of its signed variant; Theorem 49, that $c^2 = 0$ if and only if the symbol factors as $(s + a)(L_k + t)$, with Corollaries 50 and 51 and Proposition 52; Theorem 62, that no rational period-one symbol attains 14 at $k = 6$ in either matrix class; Theorem 72, that a period-one certificate covers exactly one divisibility class, which contains at most one prime, together with the identity $e_2 = -k$ for every $k \geq 3$; Theorem 65 — under the hypothesis $\det M(\zeta) \neq 0$, which an earlier version omitted — with Theorem 66 for cyclic covers; Theorems 80, 81, 82 and 83; the two refutations, Theorems 85 and 113; and Propositions 86 and 88.
- **Proved by an exact certificate.** $Z(P(n, 2)) = 6$ for every $n \geq 9$ with $n \neq 10$, and for every n divisible by 7 (Theorem 53); $Z(P(n, 3)) = 8$ for every n divisible by 10 (Theorem 58); $Z(P(n, 4)) = 10$ for every n divisible by 60, 70 or 90; $Z(P(n, 5)) = 12$ for every n divisible by 24; $Z(P(n, 7)) = 16$ for every n divisible by 120 (Theorem 61). The $k = 4, 5$ values were the first exact values for $k \geq 4$; the $k = 7$ value replaces a floating-point claim we previously made at a single n . In the $k = 2$ ($7 \mid n$), $k = 3$ and $k = 7$ cases the certificate is an integer matrix with entries in $\{0, \pm 1\}$.
- **Proved by exact certification.** $Z(P(n, 3)) = M(P(n, 3)) = 8$ for every $n \geq 17$, and $Z(P(n, 4)) = M(P(n, 4)) = 10$ for every $n \geq 29$ (Theorem 103), and $Z(P(n, 5)) = M(P(n, 5)) = 12$ for every $n \geq 162$ (Theorem 110), from Theorem 101 together with certified

identity tiles of every length in $[21, 44)$ at $k = 3$ and $[29, 62)$ at $k = 4$, plus Theorem 94 for $17 \leq n \leq 20$; at $k = 5$ the certified lengths are $[54, 73) \cup \{74, 75\}$, which do not form an interval $[L, 2L)$ but generate every $n \geq 162$ as a sum. No congruence condition appears and the primes are included. All 77 tile certifications — 23 at $k = 3$, 33 at $k = 4$, 21 at $k = 5$ — and all 17 per- n certifications are Krawczyk contraction bounds with $\alpha \leq 1.0 \times 10^{-9}$, each evaluated in *exact rational arithmetic* (Section 20); no step rests on a floating-point nullity. Each rests on a Krawczyk contraction bound for the bilinear system $A(w)K(X) = 0$, with the square subsystem chosen structurally so that it is equivalent to $AK = 0$ rather than merely a maximal-rank selection of it. By Theorem 72 no period-one certificate can reach these n at all, so this is the only route to them that we know.

- **Proved (finite).** Every tabulated value of Z , by exhaustive search over vertex subsets. Combined with the above this gives Theorem 104: $Z(P(n, k))$ is determined for *every* admissible n when $k = 2$, $k = 3$ and $k = 4$, with no value left open in any of the three rows. In particular Theorem 105, $Z(P(n, 3)) = 8$ for every $n \geq 13$ with 13 optimal: the four values $n = 13, 14, 15, 16$ are exhaustive searches and the rest is Theorem 103, so the stabilization threshold at $k = 3$ — recorded as open in [7] — is exactly 13. This is the one result here whose statement needs no matrix theory to check.
- **Numerical, not exact.** Theorem 76: $Z(P(n, 6)) = 14$ at $n = 60, 72, 84, 90, 96, 108, 120$ and $Z(P(96, 8)) = 18$. The certificates are period-three matrices whose weights come from a Newton solve, verified on the full $2n \times 2n$ matrix with a wide spectral separation and every required entry a substantial fraction of the matrix scale, but not in exact arithmetic. We do not claim these at the standard of the items above. ($Z(P(96, 7)) = 16$ was in this category and is now exact.)
- **Observation supported by computation, not proved.** The rank law $\text{rank} = \min(2d + 1, k + 1)$, verified for $3 \leq k \leq 13$ and $1 \leq d \leq 7$. Also the saturation of the reachable set of monodromies, at dimension 31 for $k = 2$ and 51 for $k = 3$.
- **Conjectural.** Conjecture 41; Conjecture 91, the leading-coefficient criterion, which is consistent with every base in the paper and proved on none beyond those; and Conjecture 97 for $k \geq 6$. For $k = 3$ and $k = 4$ the conjecture is a theorem (Theorem 103), and for $k = 5$ it is a theorem with a non-optimal threshold (Theorem 110, $N(5) \leq 162$); what remains open is whether $N(k)$ exists for every k , for which we prove only the necessary condition (3).
- **Refuted (ours).** The regular-base ceiling conjectured in Remark 89, and with it the conjecture of an earlier version that $M(B^n) = 6$ on the K_4 family of Section 21, which would have given $Z - M \rightarrow \infty$ on cubic graphs: Theorem 113 gives $M(B^n) \geq n$. The accompanying statement that the *equivariant* maximum nullity of that family is 6 was also false, because Theorem 65 needs $\det M(\zeta) \neq 0$ and a rank-one fibre makes it vanish identically. The conjecture had been written into the paper, the board and the interview script as the strongest result in the project.
- **Refuted (ours).** That the period-one construction “never exceeds 6 for every $k \geq 2$ ”. Three roots can be prescribed and no more, but the remaining $k - 2$ can land on the grid by Galois conjugacy, and do. The period-two construction we built to get past 6 was unnecessary and its values are superseded.
- **Refuted (ours).** That $k = 3$ and $k = 7$ admit no rational period-one symbol attaining $2k + 2$. That enumeration imposed $c^2 > 0$, the condition for a *symmetric* certificate, where

$c^2 \neq 0$ suffices: c^2 is the product of the two spoke weights and only its vanishing is fatal. Six symbols at $k = 3$ and one at $k = 7$ survive, giving Theorems 58 and 61.

- **Refuted (ours).** The unconditional bound $Z(B^n) \geq D$ for cyclic covers, by $Z(P(8, 2)) = 5 < 6$. Only the conditional form $Z(B^n) \geq \nu$ for an attained ν survives.
- **Refuted (ours).** The single-threshold form $n_0(k) = 5k - 2$, by $Z(P(29, 6)) = 12$; and any bound of the form $Z \leq 2(\gcd(n, j) + \gcd(n, k))$, by $Z(P(11, 3)) = 7 > 4$.
- **Refuted (ours).** That the threshold for reaching the ceiling with a general matrix is $n \geq (2k + 2)^2/5$. Spending the gauge gives (3), $n \geq (k + 1)(2k + 3)/3$, which is larger; the earlier figure led us to test $k = 6$ at $n = 27, 29, 31$, all below the true threshold of 35.
- **Refuted (ours).** That $N(k)$ in Conjecture 97 is $O(k)$. The saturation data alone suggests $4k + 3$, but (3) forces $N(k) \geq \lceil (k + 1)(2k + 3)/3 \rceil$, which is quadratic.
- **Refined (ours).** We earlier recorded as refuted the claim that the monodromy is symplectic, on the grounds that the *standard* form is not preserved. That is true and was the wrong conclusion to draw. With the docking pattern fixed there is a unique invariant skew form B , and it is nondegenerate, so tile products are symplectic with respect to B (Observation 106). The earlier statement — that the space of invariant skew forms for a single product has dimension $k + 1$, so eigenvalues come in reciprocal pairs — remains correct; it is the family, not the individual, that pins B down.
- **Observation, not proved.** Observations 106 and 107: the invariant form and the full-dimensionality of the reachable set, verified for $2 \leq k \leq 6$ but not proved. These are what reduce the existence of identity tiles for all k to a controllability question in $Sp(2k + 2, \mathbb{R})$, stated in Section 20. We do not claim that reduction settles the question: subsemigroups with nonempty interior in $Sp(2n, \mathbb{R})$ need not be the whole group.
- **Inconclusive, not negative (ours).** The per- n search of Section 20 at $k = 6$ returned nothing at $n = 41, 43, 45, 47, 49, 51$, and the tile search at $k = 5$ nothing at $41 \leq \ell \leq 45$ — the latter now explained: the first $k = 5$ tile is at $\ell = 54$, so that search was below the range where tiles exist. Both ran at a restart budget that we have since seen condemn a length that does solve: $\ell = 21$ at $k = 3$ stalled at 5.6×10^{-2} on four restarts and produced a tile at 1.4×10^{-14} on twenty-five. We therefore record these as searches we did not push far enough, not as evidence about $k = 5$ or $k = 6$. The one regularity we have found is that every length eventually solved first appeared at a residual of 10^{-2} or below, and the $k = 5$ and $k = 6$ attempts sit at 10^{-1} .
- **Withdrawn as unreliable (ours).** The eigenvalue-minimising search whose plateaus we previously reported as evidence that the ceiling is unreachable at various (n, k) . It gave maximum nullity 7 at $P(20, 3)$, where 8 is attained — by Theorem 58 and, in the symmetric class, by the method of Section 20. It collapses onto the degenerate strata (all spokes zero, or all edges zero). Its negative results should not be read as evidence, and the conclusion we drew from them, that exactness for $k \geq 6$ was unlikely, was not supported.
- **Open.** Conjecture 97 for $k \geq 6$, where we have no tiles at all and the count gives only $\ell \geq 54$. At $k = 5$ the conjecture now holds for $n \geq 162$ (Theorem 110) but not below it: we have 21 of the 54 lengths $[54, 108)$ would require, and $54 \leq n < 162$ is settled only at the 64 values that are sums of them. Also whether $M = Z$ in general, and the rank law. No

(n, k) is proved to have $M(P(n, k)) < 2k + 2$, and the one case where we expected a strict gap, $(n, k) = (10, 2)$, turned out not to be one: $M(P(10, 2)) = Z(P(10, 2)) = 6$ by the integer certificate of Remark 55.

- **Superseded (ours).** The claim that the period- d method “cannot settle prime n at all” and that a different construction would be needed there. That is correct for the period- d method and is now a theorem (Theorem 72), but it is no longer a limitation of the paper: Section 20 settles prime n for $k = 3$.
- **Not determined.** Whether Theorem 40 is attained for $k \geq 3$; whether the degeneracy of Section 19 at $n = 48, 54$ is essential.

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- [7] A. Krishnan, A correction to the zero forcing number of the generalized Petersen graphs $P(n, 3)$, arXiv:2607.19412 (2026).
- [8] A. Terras, *Zeta Functions and Chaos*, lecture notes, October 12, 2009; and *Zeta Functions of Graphs: A Stroll through the Garden*, Cambridge Studies in Advanced Mathematics 128, Cambridge University Press, 2010. The equivalence we use is stated there as Exercise 9: for a $(q + 1)$ -regular graph the Riemann Hypothesis holds if and only if every eigenvalue λ of the adjacency matrix with $|\lambda| \neq q + 1$ satisfies $|\lambda| \leq 2\sqrt{q}$.
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- [10] A. Lubotzky, R. Phillips and P. Sarnak, Ramanujan graphs, *Combinatorica* **8** (1988), 261–277.
- [11] R. Gera and P. Stănică, The spectrum of generalized Petersen graphs, *Australasian J. Combinatorics* **49** (2011), 39–45. Theorem 2.4 gives the quadratic $\delta^2 - (\alpha_j + \beta_j)\delta + \alpha_j\beta_j - 1 = 0$ with $\beta_j = 2T_k(\alpha_j/2)$, which is our Lemma 4.
- [12] T. A. Le and J. W. Sander, Integral circulant Ramanujan graphs of prime power order, *Electron. J. Combin.* **20** (2013), no. 3, #P35.

- [13] J. H. Conway and A. J. Jones, Trigonometric diophantine equations (On vanishing sums of roots of unity), *Acta Arith.* 30 (1976), 229–240.
- [14] I. Niven, *Irrational Numbers*, Carus Mathematical Monographs 11, Mathematical Association of America, 1956, Corollary 3.12.

$$Q(x, y) = 4x^2 + 4y^2 - 8\sqrt{2} |xy| + 3 \geq 0$$

one criterion, four fixed lenses, the same for every k

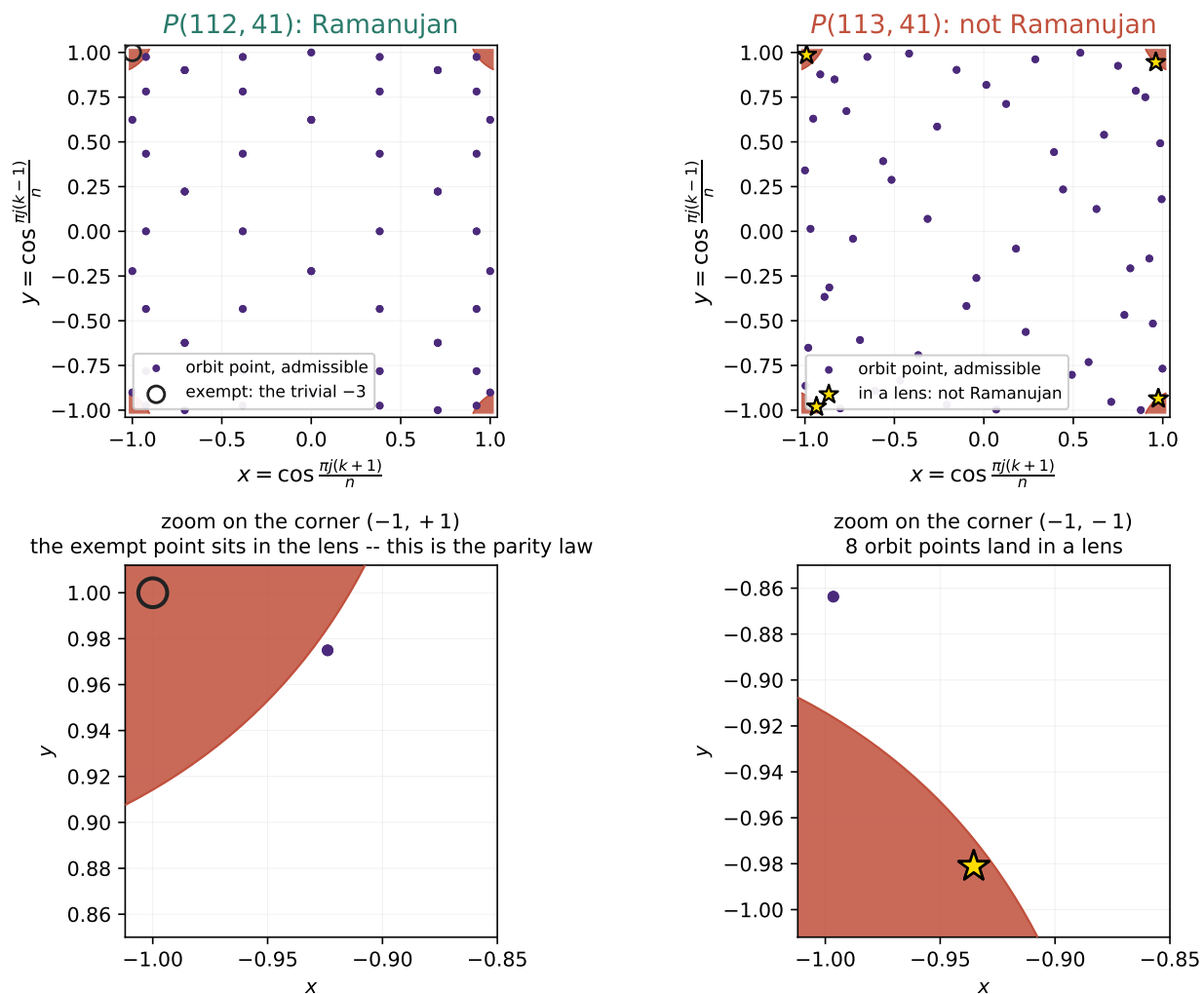


Figure 1: The corner criterion. Left, the largest Ramanujan member of the whole family; right, the cover one step larger, which fails. The four forbidden regions are the same in both pictures and for every k — only the map into the square changes. In the left-hand zoom the single point inside a region is the exempt trivial -3 , which is why bipartiteness helps.

k	B_k	count	the n for which $P(n, k)$ satisfies the RH
1	15	9	3–8, 10, 12, 14
2	23	19	5–23
3	32	18	7–16, 18, 20, 22, 24, 26, 28, 30, 32
4	42	34	9–42
5	51	28	11–26, 28, 30, 32, 34, 36, 38, 40, 42, 44, 46, 48, 50
6	61	20	13–16, 18, 20–23, 25, 27–28, 30, 32, 35, 37, 39, 42, 44, 49
7	71	39	15–36, 38, 40, 42, 44, 46, 48, 50, 52, 54, 56, 58, 60, 62, 64, 66, 68, 70
8	81	16	17, 19–22, 24, 26, 28–29, 31, 33, 35, 38, 40, 42, 49
9	91	50	19–46, 48, 50, 52, 54, 56, 58, 60, 62, 64, 66, 68, 70, 72, 74, 76, 78, 80, 82, 84, 86, 88, 90
10	101	14	21, 23, 25–26, 28, 30, 32, 34, 37, 39, 41, 43, 50, 52
11	111	36	23–26, 28–32, 35–38, 40–42, 46–48, 50, 52–53, 58, 60, 62, 64, 70, 72, 74, 82, 84, 92, 94, 96, 104, 106
12	121	9	27, 29–30, 32, 34, 38, 40, 45, 51
13	131	23	28–30, 32, 34–37, 42–44, 46, 48–49, 56, 58, 60, 70, 72, 74, 84, 86, 98
14	141	7	31, 33, 35–36, 38, 46, 53
15	151	19	33–36, 38, 40–42, 49–50, 52, 54, 56, 66, 68, 70, 82, 84, 98
16	161	6	35, 39–40, 42, 44, 52
17	171	16	37–40, 42, 44, 46–47, 56, 58, 60, 62, 74, 76, 78, 94
18	181	3	45–46, 50
19	191	17	42–44, 46, 48, 50, 52–53, 64, 66, 68, 70, 84, 86, 88, 104, 106
20	201	3	45, 49–50
21	211	11	46, 48, 50, 54, 56, 58, 70, 72, 76, 92, 96
22	221	1	49
23	231	10	52–54, 56, 60, 62, 76, 78, 84, 106
24	241	1	53
25	251	8	56, 58, 60, 66, 68, 84, 90, 92
27	271	8	60, 62, 64, 70, 72, 74, 90, 98
29	291	9	64, 66, 68, 70, 76, 78, 96, 98, 106
31	311	4	70, 74, 82, 84
33	331	5	74, 76, 78, 86, 90
35	351	5	78, 80, 84, 92, 96
37	371	3	82, 84, 88
39	391	3	88, 102, 106
41	412	2	98, 112
43	432	2	96, 98
45	452	2	100, 108

No k outside this table admits any n : the list is finite in both variables.

Table 1: The complete classification, every n and every k . For each k , B_k is the per- k bound of Theorem 12; the rows stop at $k = 45$ because by Theorem 13 no larger k admits any n at all. Generated directly from the certification run.

k	bound $2k + 2$	least n attaining it	least n from which it holds always
2	6	7	9 (it fails at $n = 8$)
3	8	10	13 (it fails at $n = 11, 12$)
4	10	18	18 (Theorem 104)
5	12	23	not determined
6	14	28	not determined; $Z(P(29, 6)) = 12$

Table 2: The bound is attained well before it is attained *always*: at $k = 3$ it first holds at $n = 10$, fails again at $n = 11$ and 12 , and holds for every $n \geq 13$ (Theorem 105). The two columns are different quantities and conflating them is easy; at $k = 6$ the bound is attained at $n = 28$ and lost at $n = 29$, so no stabilization threshold is known there. At $k = 4$ the two columns happen to coincide at 18, which is not a pattern: they differ at $k = 2$ and $k = 3$. All values by exhaustive search.



Figure 2: The deficit $\delta^*(n)$, the best achievable nontrivial spectral radius at each size, measured from the Ramanujan threshold. Green points are sizes at which an optimal member exists; the last is $n = 112$. The curve rises to the asymptote $3 - 2\sqrt{2}$, which by Proposition 33 is the most that can ever be lost. The two branches are the parity law.

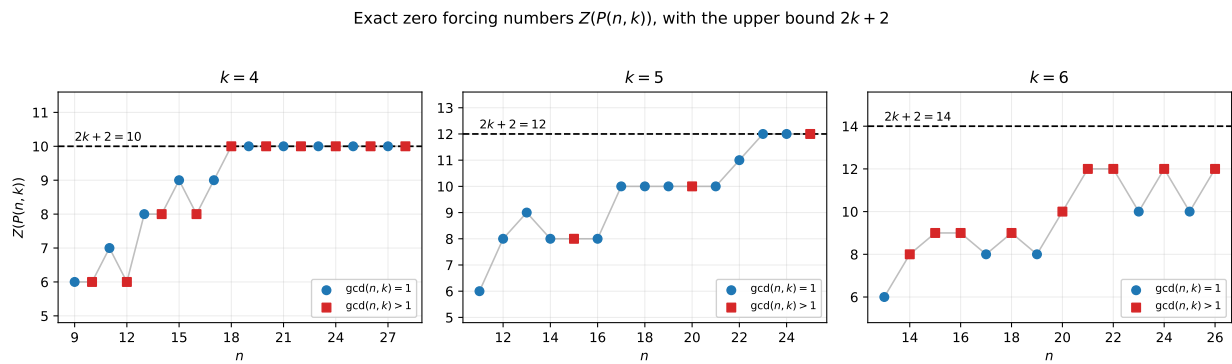


Figure 3: Exact $Z(P(n, k))$ for $k = 4, 5, 6$ against the bound $2k + 2$. Points are separated by whether $\gcd(n, k) = 1$. The sequences are non-monotone: $k = 4$ drops at $n = 12, 16$ and $k = 6$ at $n = 23, 25$.

k	$\mathcal{D}$	$\text{lcm}(\mathcal{D})$	F	a	d	c^2	nullity
2	$\{3, 10\}$	30	$sL_2 - 1$	0	0	1	$6 = 2k + 2$
2	$\{14\}$	14	Ψ_{14}	-1	0	1	$6 = 2k + 2$
3	$\{1, 4, 5\}$	20	$\Psi_1 \Psi_4 \Psi_5$	-1	-1	1	7
4	$\{4, 30\}$	60	$(s+1)L_4 - s - 2$	1	-1	1	$10 = 2k + 2$
4	$\{5, 14\}$	70	$\Psi_5 \Psi_{14}$	0	1	1	$10 = 2k + 2$
4	$\{5, 18\}$	90	$\Psi_5 \Psi_{18}$	1	0	1	$10 = 2k + 2$
5	$\{3, 6, 24\}$	24	$sL_5 - 1$	0	0	1	$12 = 2k + 2$

Table 3: Exact period-one certificates. The $k = 2$, $k = 4$ and $k = 5$ rows give $Z(P(n, k)) = M(P(n, k)) = 2k + 2$ for every n divisible by the stated modulus.

k	$2k + 2$	best period-one nullity	attained at
2	6	6	every n with $9 \leq n \leq 130$ except $n = 7, 8, 10$
3	8	8	$n = 40, 60, 80, 120$
4	10	10	$n = 60, 70, 90, 120$
5	12	12	$n = 24, 48, 72, 96, 120$
6	14	8	$n = 35, 37, 70, 74, 105, 111$
7	16	10	$n = 60, 120$
8	18	8	$n = 21, 26, 27, 39, \dots$

Table 4: The period-one construction reaches the ceiling $2k + 2$ for $k \leq 5$ and, in the computed range, falls well short for $k \geq 6$.

k	$2k + 2$	period d	$\lceil k/2 \rceil$	n	smallest required entry
6	14	3	3	60, 72	0.188, 0.162
7	16	4	4	96	0.107
8	18	4	4	96	0.260

Table 5: Period- d certificates attaining the ceiling for $k \geq 6$. The period used is in every case the one the rank law prescribes.

IV. Research Log

(May 30–September 28, 2026)

The selected dated log entries follow this page. They record the work in the order it happened rather than as a finished account, and it deliberately preserves what went wrong: conjectures that were tested and died, claims of the author's own that were later refuted, and software bugs that produced confident wrong answers before being caught.

Entries 2–4, 6–8, 12–13, 15–28 and 30–32 are not reproduced here, so entry numbers match the full log, `zf_log.pdf`, which has all 36 entries.

It is included because a reader is entitled to see which conclusions were reached first and revised later, and because the superseded versions have been left in place rather than edited away.

Investigation Log: Zero Forcing Numbers of $P(n, k)$ for $k \geq 4$

Aryan Padarthi

May 30 – September 28, 2026

Entry 1: How this problem was chosen

Date: May 30, 2026

This is the ninth research target attempted in this project. The previous eight were abandoned, and the reason matters: in every case the problem turned out to be already solved, and in every case I had inferred that it was open from the *absence* of a published result rather than from a positive statement.

- Four attempts on Turaev genus and signature bounds for positive braid closures — an area with an active professional research group.
- Periodicity of the octal game 0.45. I derived a real structural mechanism, then found the Guy–Smith periodicity theorem (1956), which settles it outright once one has computed past $n = 1018$. I had verified to 60,000.
- A “rare-index invariant” I derived for octal games, which is exactly Berlekamp’s Sparse Space Phenomenon, described in *Winning Ways* c. 1980.
- Persistent homology of linkage configuration spaces. Farber–Schütz (2007) give closed Betti-number formulas for planar polygon spaces, and Ba–Zhou (arXiv:2506.06781, 2025) prove the self-avoiding moduli spaces are diffeomorphic to Euclidean space — i.e. contractible, so persistent homology would correctly compute nothing at all.

The corrected procedure: before any investment, ask *what standard result would make this trivial*, and require a positive statement of openness. This target passes that test. Krishnan (arXiv:2607.19412, July 2026) closes with: “Determining $Z(P(n, k))$ for general k — the stabilization threshold and a rigorous lower bound for $k \geq 4$ — remains open.” That is an explicit assertion in a paper two months old, not an inference from silence.

Entry 5: A correction I had to make to my own claim

Date: June 18, 2026

After seeing $Z(P(n, 4)) = 10$ for $n = 18, \dots, 22$ — five consecutive values — I concluded the sequence had stabilized. The $k = 5$ row then showed $Z = 10$ at $n = 17, \dots, 21$, also five consecutive values, followed by 11 at $n = 22$ and 12 at $n = 23$. Five repeats are not evidence of stabilization. The $k = 4$ claim survives only because $2k + 2$ is a *proven* ceiling there, so the sequence cannot climb — but it dropped at $n = 12$ and $n = 16$, so it could in principle drop again. This is the same warning Krishnan gives about extrapolating from finite computation, arriving on my own data within an hour of my reading it.

Entry 9: A prediction, recorded before the test

Date: July 7, 2026

The thresholds at which $Z(P(n, k))$ first attains $2k + 2$ are 10, 13, 18, 23 for $k = 2, 3, 4, 5$. For $k \geq 3$ these are exactly $5k - 2$; $k = 2$ is the exception. The formula predicts $n_0(6) = 28$.

At the time of writing the $k = 6$ computation has reached $Z(P(27, 6)) = 13$, which is consistent (the ceiling 14 is not yet attained) but does not test the prediction. $P(28, 6)$ is running. This is recorded now, before the answer is known, so that a confirmation counts as a test rather than a fit — the same discipline that Entry 5 failed to apply.

Entry 10: The prediction was tested, and the conjecture died

Date: July 12, 2026

Entry 9 recorded, before the computation finished, that $n_0(6) = 5 \cdot 6 - 2 = 28$ and hence $Z(P(28, 6)) = 14$. The run took just over fourteen hours.

$Z(P(28, 6)) = 14$, with witness $\{u_0, \dots, u_{13}\}$. The narrow prediction was correct: 28 is the least n at which $k = 6$ attains its ceiling, exactly as $5k - 2$ says.

$Z(P(29, 6)) = 12$. And this refutes the conjecture. It was stated as “ $Z(P(n, k)) = 2k + 2$ for all $n \geq n_0(k)$ ” — a *stabilization* claim — and $29 > 28$ with $Z = 12 \neq 14$. So the prediction and the conjecture came apart: the thing I could check quickly was right, and the thing it was evidence for was wrong.

The cause is $\gcd(n, k)$, which had been visible in the $k = 6$ data for some time and which I twice set aside because $k = 4$ and $k = 5$ looked uniform. Sorting by $d = \gcd(n, 6)$, each class climbs on its own schedule: $d = 2$ reaches 14 at $n = 28$ while the coprime class has only reached 12 by $n = 29$. At $k = 4$ every class had already converged inside the computed range, so no separation was visible; at $k = 6$ the range finally became wide enough to see it. The apparent thresholds 10, 13, 18, 23 are where the slowest class happened to arrive, not values of a formula.

Two things worth recording about the process rather than the result. First, the prediction was useful precisely because it was written down in advance — had I fitted $5k - 2$ after seeing $Z(P(28, 6)) = 14$, I would have called it a confirmation and never computed $n = 29$. Second, this is the third time in two days that a pattern read off a short run has failed on the next data point (five repeats at $k = 5$; “ Z tracks n ” for I-graphs; now $5k - 2$). The consistent lesson is that in this family, sequences look stable well before they are.

Entry 11: A second conjecture, refuted by data I already had

Date: July 17, 2026

From 20 computed I-graphs, all satisfying $Z \leq 2(d_1 + d_2)$ with $d_1 = \gcd(n, j)$, $d_2 = \gcd(n, k)$, I wrote a conjecture into the paper asserting that bound in general. It is false, and the counterexamples were already in the validation table.

Every one of those 20 graphs had $d_1, d_2 > 1$ — that was the selection criterion for “genuinely new” I-graphs. Setting $j = 1$ gives $d_1 = 1$, and with $\gcd(n, k) = 1$ as well the bound reads $Z \leq 4$, while $Z(P(n, k)) = 2k + 2$ grows without bound. $Z(P(11, 3)) = 7$ and $Z(P(17, 4)) = 9$ refute it immediately. I had computed both values days earlier.

The error was generalising from a sample selected by a condition, to a statement about everything, without testing the boundary the condition excluded. The previous conjecture ($n_0 = 5k - 2$) at least took a fourteen-hour computation to kill; this one needed a table lookup.

What survives is narrower and still useful. The observation holds as stated for $d_1, d_2 > 1$, and a separate computation supports the underlying intuition: the minimum *bootstrapping* set for $I(18, 3, 8)$ — the object a uniform-in- n proof actually needs — has size $10 = 2(d_1 + d_2)$ against $2(j + k) = 22$, and decomposes as three consecutive vertices in each of the three outer cycles plus one inner vertex. Three consecutive vertices in an outer cycle is precisely what lets the middle one force its spoke. So the route to beating the $Z(I(n, j, k)) \leq 2(j + k)$ bound is a proof adapted to the cycle decomposition rather than to consecutive indices — but the target it proves must retain some dependence on j and k , not on d_1, d_2 alone.

Entry 14: A claim of mine, refuted — and the refutation is the result

Date: July 31, 2026

Date: September 26, 2026 **Command:** `python verification/d1_ceiling.py`

Entry 12 asserted that the period-one construction “never exceeds 6 for every k ”, because the symbol $F(s) = (s + a)L_k(s) + \alpha s + \beta$ has only three free parameters. I wrote a script to check that assertion exhaustively over all triples of grid roots, expecting it to confirm the claim. It printed 12.

The premise was right and the inference was wrong. At a root the condition is linear, so three roots determine (a, α, β) and no fourth can be imposed — but prescribing three roots *determines the other* $k - 2$, and those can land on the grid $\Gamma_n = \{2\cos(2\pi m/n)\}$ by themselves. They do so exactly when arithmetic makes them: a symbol with rational coefficients has Galois-closed roots, and the Galois orbit of $2\cos(2\pi m/n)$ stays inside Γ_n . One rational F places all $k + 1$ roots at once.

I had reasoned about how many roots I could *choose* and concluded something about how many roots could *be there*. The period-two machinery of Entry 12, built to escape a cap that was not real, is now redundant.

Entry 29: I deleted a correct citation because a summary told me to

Date: September 27, 2026

The reviewer retracted one of the eleven findings. Their W3 had said that arXiv:2607.19412 says nothing about $k \geq 4$, and that our sourcing of the $k \geq 4$ gap to that note was unsupported. I acted on it and deleted the sourcing. Then they pulled the PDF text and found the note’s final sentence:

“Determining $Z(P(n, k))$ for general k — the stabilization threshold and a rigorous lower bound for $k \geq 4$ — remains open.”

Our sourcing had been accurate and verbatim. I have restored it, having this time downloaded the PDF and read the sentence myself rather than taking anyone’s word for it, including the retraction’s.

What I want on the record is that this was my error too, not only theirs. When the report arrived I did check the claim before acting — but I checked it with a page fetch that returned the abstract plus a summarizer’s editorial remark that “the paper does not discuss $k \geq 4$.” I treated that sentence as evidence. It was not evidence; it was a guess by a tool with no access to the body, and the body is where the sentence lives. I then deleted correct text on the strength of it. The whole exchange is two parties making the same mistake in opposite directions, and the fix in both cases was ten seconds of `pdftotext`.

The rule I am taking from this: a claim about what a source does or does not say is a claim about its text, so it needs the text. Not the abstract, not a landing page, not a summary of either. I had already written a version of this rule for my own numerical results — quote the residual, not the impression — and I did not apply it to the literature.

Two consequences beyond the restoration. First, the conjecture we prove is *Conjecture 5*, numbered, and the note is sharper about it than I had conveyed: it says the content is the lower bound $Z(P(n, 3)) \geq 8$ for all $n \geq 13$, verified only to $n \leq 20$, that proving it means excluding 7-vertex forcing sets *uniformly in n* , and that “we will not assert it from finite computation, which would reproduce the gap this note corrects.” A paper that explicitly declines to extrapolate, and names uniformity in n as the missing step, is naming precisely what tiling supplies. The paper now says so and quotes it.

Second, I have applied the same standard against myself and *downgraded* a prior-art claim of my own. I had written that Duong–Kroschel–Riddell–Vander Meulen–Van Tuyl build their circulant matrices from orthogonal Hankel blocks. I have not read that paper — the publisher refused automated access three times — and that characterisation came from a summary of its abstract. So it is not a finding. The paper now cites the work as the first place a reader should look for overlap, records that we could not obtain the full text, and declines to say what is or is not in it. That is a gap in our literature review, stated as such.

And the bookkeeping, fixed at the root. The certified tile census has been 54, then 55, then 56, and $L(4)$ has been 31, 30, 29 — six numbers, six corrections, and not one of them a change to the theory. Each time I fixed the instances and each time it drifted again, because a count typed into prose is stale the moment the next tile certifies. So the

numbers are no longer typed. `verify_all.py` now reads the certificates off disk and emits `zf_numbers.tex`, which the paper and the board both `\input`. The count is now a build product. It was right to be told this three times; it was wrong of me to keep fixing instances instead of the cause.

What the suite was not checking at all. Seventy-odd checks, and not one of them checked a value of Z . Every headline claim in this project is a statement about $Z(P(n, k))$, and those numbers came out of one C program that nothing else ever tested — and the $k = 4$ values at $n = 24, \dots, 28$ depend on a rotation-symmetry reduction inside it that was likewise never tested. That is a strange blind spot to have while writing a section on how carefully the matrix side is certified. There are now cross-checks against an unreduced enumeration and against a CP-SAT integer program that does not enumerate subsets at all, over every value the paper prints, with no disagreement anywhere.

$k = 5$, by a route I had not noticed. All 21 identity tiles at $\ell = 54, \dots, 72, 74, 75$ certified, exactly. That does *not* give $[54, 108)$ — $\ell = 73$ and $76 \leq \ell \leq 107$ are still missing — so the tiling theorem does not apply as stated, and for a while I recorded $k = 5$ as still open. But $[L, 2L)$ is a sufficient condition dressed as the hypothesis; what the construction needs is only that n be a *sum* of certified lengths. The semigroup generated by those 21 lengths contains every integer $n \geq 162$. So $Z(P(n, 5)) = M(P(n, 5)) = 12$ for every $n \geq 162$, a fourth value of k , from tiles I already had. I had been reading my own hypothesis as a requirement.

Entry 33: the ceiling is not a property of the cover, and how the search lied

The question. Entry 30 left the ceiling proved for every matrix on $P(n, k)$ but only for *equivariant* matrices on a general base, and the paper recorded the gap as open: is the ceiling a property of the cover, or only of equivariant matrices on it? Closing it would have turned a fact about one family into a theorem about cyclic covers, which is why I went at it.

Two theorems first. The two-vertex elimination *chains*: on a path base with a loop at each vertex, solve the row at one end for the next fibre and repeat, each loop widening the window by $2p_t$, giving order $2 \sum p_t$ — exactly the degree span. That is Theorem ??, with the two-vertex case as $m = 2$. Cycle bases fall to something cruder: every base vertex has degree 2, so the cover is a disjoint union of $\gcd(|W|, n)$ cycles, and $M(C_N) = 2$ finishes it. Two different arguments, two different reasons.

Then the answer, which is no. The conjecture is false. A base vertex of degree 1 lifts to pendant vertices of the cover; the diagonal is free in $\mathcal{S}(G)$; so zero the pendant diagonals. The row at a pendant becomes $x_u = 0$ for its neighbour, the shared fibre dies, and what is left is one relation among $2n$ pendant coordinates. On the star-with-a-loop base, null $A = n$ against $D = 2$: the gap is not marginal, it grows without bound. Demanding minimum degree 2 does not save it either — $K_{2,3}$ gives 6 against $D = 4$. What survives is regularity, which costs nothing, since every cover this paper actually studies is cubic.

What I want to record is how I nearly missed it. I had run an adversarial search over ten connected bases and found no counterexample, and I wrote that into the paper as evidence *for* the conjecture. The star base was one of the ten. Three weaknesses compounded: Nelder–Mead in fifty dimensions, a penalty floor holding $|w| \geq 0.3$ — which excludes part of the admissible set rather than discouraging it — and an objective that summed the d smallest squared singular values, so one value at 10^{-10} hid two at 10^{-6} . Every one of those is a way of asking a weaker question than the one I thought I was asking.

What found it was not a better search. A gradient method flagged the base, and then the counterexample was written down *by hand* from the structure — pendants, free diagonal, zero — and verified by exact rank over $\mathbb{Z}$. The lesson is the one already on my own board, in the keybox about interpreting a search failure, and I walked straight past it: a failed search measures effort spent, not impossibility. The failed-search paragraph stays in the paper, relabelled. A search that does not find something is evidence about the search.

Three more of the same shape, in one day. An optimiser reported a strict gap $M(P(10, 2)) < Z$ that does not exist — two further random seeds found the nullity-6 basin. A `pgrep` wait matched its own command line and span forever; another failed to match at all, and I read a half-written file as a finished result. And a patch script died on a missing anchor inside an `&&` chain whose trailing `echo` printed “launched” anyway, so I announced a verification run that was not running, over a suite still asserting the conjecture I had just refuted. Different tools, one error: I took the success signal for the thing it was supposed to signal.

Entry 34: the obstruction, and what it is worth

Why regularity, finally. Entry 33 refuted the cover ceiling on an arbitrary base and left “regular” as a hypothesis fitted to the surviving cases, which is not a reason. The reason is one line of counting. Let S be independent and zero the diagonals on S : each row at $v \in S$ reduces to $a_{vv}x_v = 0$, which holds identically, and each row at a neighbour is one relation on $x|_S$, so vectors supported on S give $M(G) \geq |S| - |N(S)|$. On the star-with-a-loop cover that is $2n - n = n$ — exactly the nullity I had exhibited, so the bound is not merely compatible with the counterexample, it *is* the counterexample. And in a d -regular graph all $d|S|$ edges from an independent S land in $N(S)$, each vertex absorbing at most d , so $|N(S)| \geq |S|$ and the whole mechanism is unavailable. Now the hypothesis has a cause.

Then the thing I did not expect to find. I went looking for what the minimum rank community actually considers open, rather than guessing, and the Fallat–Hogben survey — which this paper already cites — asks for sufficient conditions giving $M(G) < Z(G)$ and for a parameter Y with $M \leq Y < Z$ explaining the gap. The degree span is a candidate Y , and I had been treating it as an internal bookkeeping device.

So I looked for a cover where Z exceeds it. On the K_4 base, $D = 6$ for every n while $Z = n + 2$, computed by CP-SAT with optimality proved at $n = 4, \dots, 8$. The certificate at the span came from the project’s own Galois trick pointed somewhere new: $\det M(\zeta)$ is palindromic, hence a cubic in $x = \zeta + \zeta^{-1}$, and the minimal polynomial of $2\cos\frac{2\pi}{7}$ is also a cubic, so matching them forces every coefficient equal and pins the weights to integers — edges 2, diagonals 4, 4, 5, 4, nullity exactly 6 by exact rank over $\mathbb{Z}$.

What is actually proved. The equivariant ceiling is a theorem, and the certificate attains it, so “equivariant maximum nullity is 6 while $Z = n + 2$ ” needs no conjecture: the gap is $n - 4$ and unbounded. Only the step from equivariant to all matrices is open, and that is the regular-base ceiling.

The honest size of it. I have not proved $M < Z$ on a cubic graph; I have proved that nothing rotation-invariant can close the gap, and conjectured the rest. The distinction is the whole difference between a result and a claim, and the board and the paper both state it that way. What changed is that the project now touches a question somebody else asked, which is not something I could say of it a week ago.

Entry 35: the gap family was the counterexample

1 October 2026. An outside read of the project, written from a summary and without the paper, pointed at the voltages of the K_4 base: the three edges of voltage 0 are $0-1$, $0-3$, $1-3$ — a triangle. Each fibre index carries a triangle, and the only edges leaving it go to the fibre over vertex 2. Make every triangle block rank one, $uu^\top$ with u entrywise nonzero: the pattern is respected, the diagonal is free, the rows at the $3n$ triangle vertices have rank at most $n + n$, and the rows over vertex 2 number n . So $\text{rank } A \leq 3n$ and $\text{null } A \geq n$. Exact rank over $\mathbb{Q}$ gives nullity exactly n for $n = 4, \dots, 10$ (verification/k4_rank_one.py).

What that kills. The conjecture $M(B^n) = 6$, and with it the cubic family with $Z - M \rightarrow \infty$ that Entry 34 called the strongest result in the project. $M \geq n$ and $Z = n + 2$, so the gap is at most 2. It kills the regular-base ceiling, since K_4 is cubic. And it kills a

sentence I had called unconditional, that the *equivariant* maximum nullity is 6: the rank-one matrix is equivariant and $\det M(\zeta) \equiv 0$ for it, so every block is singular. `thm:cover` needed the hypothesis $\det M(\zeta) \not\equiv 0$ and did not state it.

Why both attacks missed it. The diagonal-zeroing attack varies diagonals only; the gradient search starts from generic weights. The counterexample lives where three of the four fibre blocks are singular, which neither reaches. Entry 33 said a failed search is evidence about the search. This is the second conjecture the same lesson has cost.

What replaces regularity. The top coefficient of $\det M(\zeta)$, computed symbolically on every base in the paper. Where the ceiling is proved it is a monomial in edge weights alone: $w_1 w_k$ on the two-loop base, the product of the loop weights on a path with loops, the product of all edge weights on a cycle. On each of the three bases where the ceiling fails it contains a free diagonal: $-w_{12} w_{23} (w_{01} w_{03} - d_0 w_{13})$ on K_4 . That is the hypothesis every proof actually used, and it is now `conj:leading`. Regularity was a hypothesis fitted to the cases that survived two weak attacks; the general obstruction is $M(G) \geq |V| - \text{mr}(G[S]) - 2|V \setminus S|$ for any S , of which the independent-set bound of Entry 34 is the case $\text{mr}(G[S]) = 0$, and regularity does nothing to $\text{mr}(G[S])$.

Also corrected today. The abstract, the board and the arXiv draft said $Z = M = 2k + 2$ “for all n at $k = 2, 3, 4$ ”. Krishnan’s table has $Z(P(11, 3)) = Z(P(12, 3)) = 7$ and the paper itself has $Z(P(8, 2)) = 5$. The theorems carry the right thresholds; the summaries did not.

Found the same day. A pull request merged on 26 September 2026 in an automated proof repository claims a Lean-checked proof of Krishnan’s Conjecture 5 by an isoperimetric argument on forcing sets. It is not a paper and nobody has reviewed it, but it is dated, and nothing of mine is public.

References

- [1] S. Rashidi, N. Shajareh Poursalavati, M. Tavakkoli, *J. Algebra Combin. Discrete Struct. Appl.* 7(2) (2020), 183–193.
- [2] A. Krishnan, arXiv:2607.19412 (2026).

Entry 36: making the mathematics the spine

3 October 2026. [AI-DRAFTED — rewrite in your own words before this log is shown to anyone.] The question I set myself: how much of this project is mathematics and how much is computation, and can the line be moved. Three uses of the computer, three different statuses. The finite census and the small values of Z are finite case checks in exact arithmetic after a theorem has made the problem finite; nobody objects to that. The existence of the tiles is a computer-assisted existence proof: Krawczyk’s fixed-point theorem with its hypothesis verified in exact rational arithmetic, the four-colour kind of proof. The verification suite is bookkeeping and not mathematics at all, and I should stop presenting it as if it were evidence.

What moved. The closed form for $k \leq 9$ had been a proposition “verified case by case”. It is now a theorem: the negative set of D_k on $[-1, 1]$ is the end band(s) alone exactly for $k \in \{1, 2, 3, 4, 5, 7, 9\}$, by Sturm’s theorem (an integer computation on the coefficients, exact root isolation over $\mathbb{Q}$ in `verification/interior_bands.py`), and then only the grid point $j = 1$ — and for odd n and odd k only $j = (n - 1)/2$ — can enter a band. For $k = 2$ the whole argument fits on a page. From $k = 10$ on there is always an interior band, which is why the census is needed there.

A second theorem I should have seen earlier. The paper’s rotation-invariant certificates ($10 \mid n$ at $k = 3$; $60, 70, 90 \mid n$ at $k = 4$; $24 \mid n$ at $k = 5$) were constructions, with no claim that they were all of them. They can be classified. A rational symbol has a Galois-closed root set, a union of orbits $O_d = \{2 \cos(2\pi l/d) : \gcd(l, d) = 1\}$ of total size $k + 1$; the monic product of the minimal polynomials must equal $(s + a)(t_k(s) + d') - c_0$, which forces every middle coefficient; and realisability is $c_0 \neq 0$. That is a finite enumeration (`verification/period_one_classification.py`), and it gives an *if and only if*: a rotation-invariant matrix with rational symbol reaches nullity $2k + 2$ on $P(n, k)$ exactly when n is divisible by one of 7, 9, 15, 20 ($k = 2$); 10, 24, 42 ($k = 3$); 60, 70, 90 ($k = 4$); 24, 126 ($k = 5$); and never for $k = 6$, which recovers the $k = 6$ impossibility as a corollary. The classes 24 and 42 at $k = 3$ and 126 at $k = 5$ are new.

Still not pure. Tile existence. The paper already proves why forts, the fort LP and spectral bounds cannot give the all- n lower bound; a hand proof needs a new idea. Two candidates are written down in the handoff.

V. Verification Output

(September 28, 2026)

One script re-derives every computational claim in the paper from scratch and reports pass or fail per check. Its current output follows, unedited.

```

1. Realisability (thm:realis):  $c^2 = 0$  iff  $F = (s+a)(L_k + t)$ 
[PASS] k=2
[PASS] k=3
[PASS] k=4
[PASS] k=5
[PASS] k=6
[PASS] k=7
[PASS] k=8
2. Antipodal corollary:  $c^2 = 0$  identically for even  $k$ , not for odd
[PASS] k=2 (even)
[PASS] k=3 (odd)
[PASS] k=4 (even)
[PASS] k=5 (odd)
[PASS] k=6 (even)
[PASS] k=7 (odd)
[PASS] k=8 (even)
3. k=2 closed form:  $c^2 = -(s_1+s_2)(s_1+s_3)(s_2+s_3)$ 
[PASS] identity
4.  $e_2 = -k$  for every period-one symbol with  $k \geq 3$  (thm:congbound)
[PASS] k=3
[PASS] k=4
[PASS] k=5
[PASS] k=6
[PASS] k=7
[PASS] k=8
[PASS] k=9
[PASS] k=10
5. Integer certificates: the signed adjacency matrix
[PASS] signed Adj P(10,3) nullity 8 --- got 8
[PASS] signed Adj P(20,3) nullity 8 --- got 8
[PASS] signed Adj P(30,3) nullity 8 --- got 8
[PASS] signed Adj P(120,7) nullity 16 --- got 16
[PASS] signed Adj P(30,2) nullity 6 --- got 6
[PASS] signed Adj P(60,2) nullity 6 --- got 6
[PASS] Psi_7 certificate P(7,2) nullity 6 --- got 6
[PASS] Psi_7 certificate P(14,2) nullity 6 --- got 6
[PASS] Psi_7 certificate P(21,2) nullity 6 --- got 6
[PASS] Psi_7 certificate P(28,2) nullity 6 --- got 6
6. k=2 explicit certificate at the three most negative grid values
( $c^2 > 0$  is checked for all  $n$ ; the nullity is checked two ways, by
counting grid roots of the symbol exactly and, for moderate  $n$  only,
by singular values -- the three roots crowd within  $O(1/n^2)$  of  $-2$ ,
so the matrix is exact but numerically stiff for large  $n$ )
[PASS]  $c^2 > 0$  for every  $n \geq 9$ ,  $n \neq 10$  (up to 1001) --- checked 9..199, 500, 1001
[PASS] three prescribed roots are distinct and interior --- checked 9..199, 500, 1001, 5000
[PASS] P(9,2) nullity 6 numerically
[PASS] P(11,2) nullity 6 numerically
[PASS] P(20,2) nullity 6 numerically
[PASS] P(39,2) nullity 6 numerically
[PASS] P(60,2) nullity 6 numerically
7. Saved interval-certified matrices re-evaluated (k=3)
[PASS] P(17,3) nullity 8, edges nonzero --- nullity 8, min edge 0.1462
[PASS] P(18,3) nullity 8, edges nonzero --- nullity 8, min edge 0.2058

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[PASS] P(19,3) nullity 8, edges nonzero --- nullity 8, min edge 0.0145
[PASS] P(20,3) nullity 8, edges nonzero --- nullity 8, min edge 0.1424
[PASS] P(21,3) nullity 8, edges nonzero --- nullity 8, min edge 0.3059
[PASS] P(22,3) nullity 8, edges nonzero --- nullity 8, min edge 0.0878
[PASS] P(23,3) nullity 8, edges nonzero --- nullity 8, min edge 0.0647
[PASS] P(24,3) nullity 8, edges nonzero --- nullity 8, min edge 0.0405
[PASS] P(25,3) nullity 8, edges nonzero --- nullity 8, min edge 0.1584
[PASS] P(26,3) nullity 8, edges nonzero --- nullity 8, min edge 0.0859
[PASS] P(27,3) nullity 8, edges nonzero --- nullity 8, min edge 0.2439
[PASS] P(28,3) nullity 8, edges nonzero --- nullity 8, min edge 0.0296
[PASS] P(29,3) nullity 8, edges nonzero --- nullity 8, min edge 0.2612
[PASS] P(30,3) nullity 8, edges nonzero --- nullity 8, min edge 0.3665
[PASS] P(31,3) nullity 8, edges nonzero --- nullity 8, min edge 0.2290
[PASS] P(32,3) nullity 8, edges nonzero --- nullity 8, min edge 0.0445
[PASS] P(33,3) nullity 8, edges nonzero --- nullity 8, min edge 0.2802
  (17 saved certificates re-evaluated)
[PASS] thm:p17: a saved certificate exists for every n in [17,33] --- all 17 present
8. k=3 identity tiles and their compositions
[PASS] tiles cover a full interval [L,2L] --- lengths 21..43 (23 tiles)
[PASS] single tiles give nullity 8 --- all 23 lengths
[PASS] multi-tile concatenations give nullity 8 --- checked n up to 55
9. Certified tiles: nullity, docking exactness, coverage
[PASS] k=3: every n >= 21 is a sum of certified lengths --- 23 tiles, lengths 21..43 (= [L,2L], so L itself)
[PASS] k=3: every certified tile has nullity exactly 8 --- all 23 tiles
[PASS] k=3: docking weights exact in every certified tile --- all tiles, to 1e-13
[PASS] k=4: every n >= 29 is a sum of certified lengths --- 33 tiles, lengths 29..61 (= [L,2L], so L itself)
[PASS] k=4: every certified tile has nullity exactly 10 --- all 33 tiles
[PASS] k=4: docking weights exact in every certified tile --- all tiles, to 1e-13
[PASS] k=5: every n >= 162 is a sum of certified lengths --- 21 tiles, lengths 54..75; 44 smaller n not covered
[PASS] k=5: every certified tile has nullity exactly 12 --- all 21 tiles
[PASS] k=5: docking weights exact in every certified tile --- all tiles, to 1e-13
10. The paper's own numbers against the computed ones
[PASS] paper's certification count matches the 56 certified tiles --- all say 56
[PASS] Status section names the same k as thm:complete --- both say [2, 3, 4]
[PASS] k=3: paper's L matches the least certified tile length (21) --- L=21
[PASS] k=4: paper's L matches the least certified tile length (29) --- L=29
11. The Krawczyk test itself, re-run in EXACT arithmetic
[PASS] tile 21 at k=3: exact Krawczyk contraction --- alpha = 2.154e-10 < 1 exactly
[PASS] tile 29 at k=4: exact Krawczyk contraction --- alpha = 1.812e-10 < 1 exactly
[PASS] gn 17 at k=3: exact Krawczyk contraction --- alpha = 2.128e-11 < 1 exactly
12. Z values cross-checked against independent solvers
[PASS] Z(P(12,3)) = 7, by every method that completed --- 3/3 completed: reduced 7, unreduced 7, CP-SAT 7
[PASS] Z(P(8,2)) = 5, by every method that completed --- 3/3 completed: reduced 5, unreduced 5, CP-SAT 5
[PASS] Z(P(12,4)) = 6, by every method that completed --- 3/3 completed: reduced 6, unreduced 6, CP-SAT 6
[PASS] recorded cross-validation sweep has no disagreement --- 79 rows agree
13. The graph Riemann Hypothesis: classification of Ramanujan P(n,k)
[PASS] block decomposition reproduces the full spectrum --- max deviation 1.8e-14 over k=1..10, n<=60
[PASS] spectra match published values for Petersen and Desargues --- P(5,2) -> {3,1,-2}, P(10,3) -> {+-3,+-2,+-1}
[PASS] criterion G_k <= 7 agrees with the spectral test --- k=1..10, n<=260
[PASS] D_k(1)=-7, D_k(0)=17, D_k(-1)=-7 (k odd) / 9 (k even), deg 2k+2 --- k=1..12; these give the bad band at
[PASS] side condition 7 + 4u T_k(u) stays positive on [-1,1] --- minimum over k=1..12 is 3.000, and |4u T_k| <=
[PASS] finiteness bound n <= 2 pi / arccos(u_k) holds for every k --- k=1..11, attained exactly at 3 of them
[PASS] parity bound n <= pi / arccos(-v_k) holds for odd k, odd n --- k=1,3,...,11, attained exactly at 4 of them
[PASS] uniform lemma: G_k > 7 on the whole interval the proof forbids --- min G = 7.004872 at the band at 0, 7.0
[PASS] uniform bound n <= pi(2+sqrt(2)) sqrt(k^2+1), halved for odd k odd n --- holds against the certified class
[PASS] every case below the bound decided in exact arithmetic --- 458 cases certified, 0 disagreements with the
[PASS] the classification is complete for k <= 10 --- 247 Ramanujan graphs; largest n per k: 1:14, 2:23, 3:32, 4:41
14. Reduction on every two-vertex cover (thm:red2)
[PASS] thm:red2: elimination gives the predicted offsets, ends nonzero --- 8 bases, incl. p>1 and w!=0 which thm

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[PASS] thm:red2: $x \rightarrow (x, Lx)$ is a bijection onto $\ker A$, and $\text{null } A \leq 2(p+q)$ --- $\text{null } A = \text{null } R$, lift residual

[PASS] thm:theta: $\text{null } A = \dim \ker(T-I) \leq D$ on theta covers --- 6 bases incl. four-edge; state dimension equals

[PASS] thm:order: $\text{order} = p+q+\max(p, t_m)+\max(q, t_m)$ --- 12 bases, measured vs counted

[PASS] thm:order: $\text{order} = D$ iff $t_m \leq \min(p, q)$ --- the characterisation holds on every base tested

15. Slice lemma (lem:slice)

[PASS] spokes $\rightarrow +1$ and $|\text{outer}| \rightarrow 1$, nullity preserved, both parities --- max deviation $2.7e-15$

16. The corner criterion, and the classification for every k

[PASS] both factors of D_k collapse to $Q(x, y)$, independent of k --- max deviation $4.1e-14$ over $k=1..14$

[PASS] corner criterion agrees with the spectral test --- 1512 cases, $k=1..12$

[PASS] lens area $3/2 - \sqrt{2} + (3/8)\log((1+\sqrt{2})/3)$ matches the integral --- 0.004321924509 of the square, i.e.

[PASS] Dirichlet witness forces failure for every $n \geq$ threshold, every k --- threshold $n \geq 231$; worst value 0.

[PASS] the classification is finite in BOTH variables --- 460 pairs, 324 isomorphism classes; largest $n=112$, lar

[PASS] no case in the region is decided on a margin float64 could not see --- smallest $|\text{margin}|$ is $6.11e-05$, at

[PASS] the all- k census reproduces the exactly certified $k \leq 10$ list --- 247 graphs at $k \leq 10$, all recovered

17. Structure of the classification: locality, gcd safety (sec:closed)

[PASS] thm:local: Ramanujan iff no divisor $m \geq 3$ of n is bad for k --- 8280 cases against the direct criterion,

[PASS] cor:divclosed: the Ramanujan set of each k is divisor-closed --- 202 (n, d) pairs with $d \mid n$, $d > 2k$

[PASS] lem:mingcd: min over units of $||jc/m||$ equals $\gcd(c, m)/m$ --- $m < 120$, $c < 200$; the identity the safety th

[PASS] thm:gcfsafe: $m/\gcd(k+1, m)$ in $\{2..7\}$ implies m is safe --- fired on 1576 of 11820 pairs, never on a bad l

[PASS] thm:closedsmall: exact where claimed, and fails where we say it does --- holds for $k=[1, 2, 3, 4, 5, 7, 9]$

[PASS] lem:bands: D_k has end bands only for k in $\{1, 2, 3, 4, 5, 7, 9\}$ (exact Sturm root isolation, $k \leq 45$) --- end ba

[PASS] lem:bands: D_k is an even polynomial exactly for odd k --- $k \leq 45$

18. The Riemann Hypothesis on every cubic cyclic cover (thm:covcorner)

[PASS] character blocks reproduce the spectrum on both two-vertex bases --- max deviation $8.7e-15$

[PASS] the two-loop spectrum does not depend on the edge voltage c --- max deviation $3.3e-15$ over every c

[PASS] corner criterion decides every two-loop cover --- 531 covers, brute-forced from the full adjacency matrix

[PASS] theta criterion decides every cyclic Haar cover --- 384 covers, brute-forced from the full adjacency matr

[PASS] both uniform bounds hold on every Ramanujan cover found --- 485 Ramanujan covers, none above its bound

[PASS] no cover of ANY fixed cubic base is Ramanujan past its threshold --- $\text{theta}(0, 1, 2)$: max $n=15 < 156$; $\text{theta}(a$

19. The cover ceiling: two theorems and a refutation

[PASS] thm:redpath: degree span of $\det M(\zeta)$ is $2 \sum(p_t)$ --- 8 path-with-loops bases, $m = 2$ to 4 , checked sy

[PASS] thm:redpath: non-equivariant weights cannot beat $2 \sum(p_t)$ --- $p=(1, 2)$ $n=9$ $D=6$; $p=(1, 3)$ $n=11$ $D=8$; $p=(2$

[PASS] thm:redcycle: degree span is $2|W|$ with W the holonomy --- 7 cycle bases, checked symbolically

[PASS] thm:redcycle: the cover is 2-regular with $\gcd(|W|, n)$ components --- 42 (base, n) pairs -- this is why $M(C$

[PASS] thm:cexcover: star-with-a-loop breaks the ceiling, $\text{null } A = n$ vs $D = 2$ --- 9 (n, weight) pairs, ranks com

[PASS] rem:mindeg: minimum degree 2 does not repair it $(K_{\{2,3\}}, D=4)$ --- $n=5$: $\text{null}=6$ vs $D=4$; $n=7$: $\text{null}=8$ vs $D=4$

[PASS] the diagonal-zeroing attack alone never exceeds D on a regular base --- 10 (regular base, n) pairs, exact

[PASS] thm:k4rankone: rank-one triangle blocks give $\text{null } A = n$ on the K_4 cover, against $D = 6$ (regular-base ceiling)

[PASS] thm:cover needs $\det M(\zeta) \neq 0$: the rank-one equivariant matrix has $\det M(\zeta) = 0$ identically --- ev

[PASS] conj:leading: top coefficient of $\det M(\zeta)$ is a monomial in edge weights on every base where the ceiling

[PASS] conj:leading: top coefficient contains a free diagonal on every base where the ceiling fails --- star-4 +

20. The independent-set obstruction, and the cubic gap family

[PASS] cor:regblocks: d -regular implies $|N(S)| \geq |S|$ for independent S --- 2798 (regular graph, independent set

[PASS] sec:gapfamily: the K_4 covers are cubic, connected, with GENERIC $D = 6$ for all n --- $n = 4..10$, $|V| = 16..$

[PASS] prop:k4cert: the integer weights realise $16(z^7-1)/(z-1)$ --- coefficients of $z^3 \det M$ all equal 16

[PASS] prop:k4cert: exact nullity over Z equals the degree span --- $n=7$: $\text{null}=6=D$; $n=14$: $\text{null}=6=D$

[PASS] prop:k4z: $Z = n+2$ by CP-SAT with optimality proved --- $n=4$: $Z=6$; $n=5$: $Z=7$; $n=6$: $Z=8$; $n=7$: $Z=9$; $n=8$: $Z=10$

21. The deficit (prop:deficit)

[PASS] $\delta(n, k) < 3 - 2\sqrt{2}$ for every n , and δ^* rises towards it --- max $\delta^* = 0.129473 < \tau = 0.171$

22. $M(P(10, 2))$ exactly, and the period-one ceiling

[PASS] $M(P(10, 2))=6$: a period-two solution over Q exists --- $a=1/5$, $c=1$, $e=1$, $f=-5$, $s=-1$, $t=-15$

[PASS] $M(P(10, 2))=6$: the witness carries exactly the $P(10, 2)$ pattern --- 20×20 , symmetric=True, missing=0, extra

[PASS] $M(P(10, 2))=6$: exact nullity over Q equals 6 --- exact rank 14, nullity 6; with $M \leq Z = 6$ this gives $M(P($

[PASS] no period-one matrix attains nullity 6 at $P(10, 2)$ --- 10 block-class subsets of total dimension 6, 0 admi

[PASS] period-one ceiling explains both missing certificates --- $P(10, 2)$: bound $5 < 6$; $P(24, 4)$: bound $8 < 10$ --

[PASS] CONTROL: the period-one bound contradicts no existing certificate --- 16 certified (n, k) checked, 0 viola

[PASS] the period-one bound reproduces the paper's two exceptions at $k=2$ --- blocked n at $k=2$: $[8, 10]$ -- exactl

23. Emit the manuscript's numbers from the certificates on disk

23. Classification of rational-symbol rotation-invariant certificates (thm:ratclass)

[PASS] thm:ratclass: the admissible classes are 7,9,15,20 | 10,24,42 | 60,70,90 | 24,126 | none | 120 for k=2..7
[PASS] thm:ratclass: every listed certificate has nullity $2k+2$ on its class and 0 off it --- 26 certificates, k=
[PASS] manuscript numbers emitted from the certificates on disk --- 56 tiles at k=3,4 + 21 at k=5; L(3)=21, L(4)=

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ALL CHECKS PASSED

145 checks in 332.6 s (5.5 min) on 10 cores

SUITE_DONE

VI. A Certification Transcript

(September 27, 2026)

What “interval-certified” means in practice, for one family of tiles. Each entry is an independent Krawczyk contraction proof with its own unknown count, matrix norm and residual; nothing is shared between them.

certifying 22 identity tiles at $k=3$

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1=22: 398 unknowns, 352 equations, 324 structural, 28 dependent; 40 docking pins, 74 fixed total
      structural 3.11e-61, dependent 2.72e-61, ||Y||=5.39e+02, alpha=6.70e-11, nullity 8, docking exact True, min|edge| 0.1500 CEF
1=23: 419 unknowns, 368 equations, 340 structural, 28 dependent; 40 docking pins, 79 fixed total
      structural 3.11e-61, dependent 4.67e-61, ||Y||=7.55e+02, alpha=9.25e-11, nullity 8, docking exact True, min|edge| 0.1500 CEF
1=24: 440 unknowns, 384 equations, 356 structural, 28 dependent; 40 docking pins, 84 fixed total
      structural 6.22e-61, dependent 1.94e-61, ||Y||=2.48e+03, alpha=3.24e-10, nullity 8, docking exact True, min|edge| 0.1500 CEF
1=25: 461 unknowns, 400 equations, 372 structural, 28 dependent; 40 docking pins, 89 fixed total
      structural 6.22e-61, dependent 2.24e-61, ||Y||=9.96e+02, alpha=1.23e-10, nullity 8, docking exact True, min|edge| 0.1500 CEF
1=26: 482 unknowns, 416 equations, 388 structural, 28 dependent; 40 docking pins, 94 fixed total
      structural 3.5e-61, dependent 1.94e-61, ||Y||=5.16e+02, alpha=6.88e-11, nullity 8, docking exact True, min|edge| 0.1500 CEF
1=27: 503 unknowns, 432 equations, 404 structural, 28 dependent; 40 docking pins, 99 fixed total
      structural 6.22e-61, dependent 1.56e-61, ||Y||=1.51e+03, alpha=1.89e-10, nullity 8, docking exact True, min|edge| 0.1500 CEF
1=28: 524 unknowns, 448 equations, 420 structural, 28 dependent; 40 docking pins, 104 fixed total
      structural 6.22e-61, dependent 2.48e-61, ||Y||=1.04e+03, alpha=1.31e-10, nullity 8, docking exact True, min|edge| 0.1500 CEF
1=29: 545 unknowns, 464 equations, 436 structural, 28 dependent; 40 docking pins, 109 fixed total
      structural 3.31e-61, dependent 1.56e-61, ||Y||=3.84e+02, alpha=4.77e-11, nullity 8, docking exact True, min|edge| 0.1500 CEF
1=30: 566 unknowns, 480 equations, 452 structural, 28 dependent; 40 docking pins, 114 fixed total
      structural 6.22e-61, dependent 3.69e-61, ||Y||=5.86e+02, alpha=7.43e-11, nullity 8, docking exact True, min|edge| 0.1500 CEF
1=31: 587 unknowns, 496 equations, 468 structural, 28 dependent; 40 docking pins, 119 fixed total
      structural 4.67e-61, dependent 6.51e-61, ||Y||=6.67e+02, alpha=8.49e-11, nullity 8, docking exact True, min|edge| 0.1500 CEF
1=32: 608 unknowns, 512 equations, 484 structural, 28 dependent; 40 docking pins, 124 fixed total
      structural 3.11e-61, dependent 2.14e-61, ||Y||=6.86e+02, alpha=8.71e-11, nullity 8, docking exact True, min|edge| 0.1500 CEF
1=33: 629 unknowns, 528 equations, 500 structural, 28 dependent; 40 docking pins, 129 fixed total
      structural 2.53e-61, dependent 2.33e-61, ||Y||=5.64e+02, alpha=7.14e-11, nullity 8, docking exact True, min|edge| 0.1500 CEF
1=34: 650 unknowns, 544 equations, 516 structural, 28 dependent; 40 docking pins, 134 fixed total
      structural 4.76e-61, dependent 4.08e-61, ||Y||=2.88e+02, alpha=3.66e-11, nullity 8, docking exact True, min|edge| 0.1500 CEF
1=35: 671 unknowns, 560 equations, 532 structural, 28 dependent; 40 docking pins, 139 fixed total
      structural 3.11e-61, dependent 5.25e-61, ||Y||=1.43e+03, alpha=1.77e-10, nullity 8, docking exact True, min|edge| 0.1500 CEF
1=36: 692 unknowns, 576 equations, 548 structural, 28 dependent; 40 docking pins, 144 fixed total
      structural 7.0e-61, dependent 3.11e-61, ||Y||=2.17e+03, alpha=2.79e-10, nullity 8, docking exact True, min|edge| 0.1500 CEF
1=37: 713 unknowns, 592 equations, 564 structural, 28 dependent; 40 docking pins, 149 fixed total
      structural 3.11e-61, dependent 3.4e-61, ||Y||=8.29e+02, alpha=1.06e-10, nullity 8, docking exact True, min|edge| 0.1500 CEF
1=38: 734 unknowns, 608 equations, 580 structural, 28 dependent; 40 docking pins, 154 fixed total
      structural 3.11e-61, dependent 2.14e-61, ||Y||=8.16e+02, alpha=1.01e-10, nullity 8, docking exact True, min|edge| 0.1502 CEF
1=39: 755 unknowns, 624 equations, 596 structural, 28 dependent; 40 docking pins, 159 fixed total
      structural 3.11e-61, dependent 4.67e-61, ||Y||=5.92e+02, alpha=7.70e-11, nullity 8, docking exact True, min|edge| 0.1500 CEF
1=40: 776 unknowns, 640 equations, 612 structural, 28 dependent; 40 docking pins, 164 fixed total
      structural 3.69e-61, dependent 1.94e-61, ||Y||=1.15e+03, alpha=1.43e-10, nullity 8, docking exact True, min|edge| 0.1500 CEF
1=41: 797 unknowns, 656 equations, 628 structural, 28 dependent; 40 docking pins, 169 fixed total
      structural 6.22e-61, dependent 5.45e-61, ||Y||=8.81e+02, alpha=1.11e-10, nullity 8, docking exact True, min|edge| 0.1500 CEF
1=42: 818 unknowns, 672 equations, 644 structural, 28 dependent; 40 docking pins, 174 fixed total
      structural 6.22e-61, dependent 3.11e-61, ||Y||=1.65e+03, alpha=2.12e-10, nullity 8, docking exact True, min|edge| 0.1500 CEF
1=43: 839 unknowns, 688 equations, 660 structural, 28 dependent; 40 docking pins, 179 fixed total

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